ZORK I

THE GREAT UNDERGROUND EMPIRE

A maximum-score route to the Stone Barrow ending

SCORE	MOVES	DEATHS	INSTRUCTIONS
350 / 350	**400**	**0**	**417,780**

Verified story: Z-machine V3, Release 88, Serial 840726. The route reaches the Stone Barrow ending with 350 points and no deaths. It was replayed from a fresh game with deterministic seed 1.

How to use this guide

Follow the numbered stages in order. Commands are shown in Courier. Directions assume the route has left you in the stated room. At the few randomized combat encounters, repeat the stated attack until the foe is gone; the supplied regression route records one deterministic successful run.

Route checkpoints

The exact local replay consumes 405 player command inputs and 417,780 VM instructions. The final score is 350/350 at the Living Room trophy case; entering the Stone Barrow completes the game. The replay contains additional state checks and repeated combat inputs; the numbered steps below condense those into a usable walkthrough.

Story fingerprint: SHA-256 158F1F63B1302591BBCE30E1EC23B17909D2A66E39403ED14593A75280B1E7F9

The accompanying opcode inventory uses 1-based instruction sequence numbers and hexadecimal byte addresses.

01. The house, painting, and troll

Read the opening leaflet if desired, then enter through the open window. The painting is the first trophy. The lantern and rope are essential for the underground route.

```
open mailbox
read leaflet
drop leaflet
go south
go east
open window
enter house
go west
take lamp
move rug
open trap door
turn on lamp
go down
go south
go east
take painting
go north
go up chimney
go up
take knife
take rope
go down
go west
open case
put painting inside case
drop knife
take sword
open trap door
go down
go north
kill troll with sword (repeat until dead)
drop sword
go east
go east
go southeast
go east
tie rope to railing
go down
```

02. The coffin and rainbow treasures

From the Torch Room, collect the coffin in the Egyptian Room. The altar prayer returns you to the forest. Bring the coffin to the End of the Rainbow; its contents and the sceptre reveal another treasure.

```
go south
go east
take coffin
```

```
go west
go south
pray
turn off lamp
go south
go north
go east
go down
go down
go north
drop coffin
open coffin
take sceptre
wave sceptre
take gold
take coffin
go southwest
go up
go up
go northwest
go west
enter house
open bag
take garlic
go west
put coffin in case
put gold in case
put sceptre in case
```

03. Lower the dam and exorcise Hades

Open the dam from its maintenance room. The yellow control and wrench-operated bolt lower the lake. From Deep Canyon use the southwest route to the Dome Room, avoiding the Loud Room until it is quiet. Take the bell, book, and candles to the lower temple and perform the rite.

```
open trap door
turn on lamp
go down
go north
go east
go north
go northeast
go east
go north
take matches
go north
take wrench
take screwdriver
push yellow button
go south
go south
turn bolt with wrench
```

```
drop wrench
go south
go southwest
go south
go southeast
go east
climb down rope
take torch
turn off lamp
go south
take bell
go south
take candles
take book
go down
go down
ring bell
take candles
light match
light candles with match
read book
drop book
go south
take skull
go north
go up
go north
put out candles
rub mirror
go north
go west
go north
go west
go north
go east
```

04. Coal mine, diamond, and jade

Use the basket to move the torch, coal, screwdriver, and diamond between mine levels. The gas-room bracelet is collected later through the mirror branch. Carry garlic through the Bat Room.

```
put torch in basket
put screwdriver in basket
turn on lamp
go north
go down
go east
go northeast
go southeast
go southwest
go down
go down
```

```
go south
take coal
go north
go up
go up
go north
go east
go south
go north
go up
go south
put coal in basket
lower basket
go north
go down
go east
go northeast
go southeast
go southwest
go down
go down
go west
drop all
go west
take coal
take screwdriver
take torch
go south
open lid
put coal in machine
close lid
turn switch with screwdriver
drop screwdriver
open lid
take diamond
go north
put torch in basket
put diamond in basket
go east
take skull
take lamp
take garlic
go east
go up
go up
go north
go east
go south
go north
go up
go south
```

```
raise basket
take diamond
take torch
turn off lamp
go west
take jade
go south
go east
go south
go down
go up
put jade in case
put diamond in case
put torch in case
put skull in case
```

05. Reservoir, boat, emerald, and scarab

The opened dam exposes the trunk, pump, and trident. On the river, wait until the warning buoy is visible before taking it and steering east to the sandy landing. Drop the buoy to make room for the shovel; dig until the scarab appears.

```
turn on lamp
go down
go north
go east
go north
go northeast
go north
take trunk
go north
take pump
go north
take trident
go south
go south
go south
go east
go east
inflate plastic with pump
drop pump
go inside boat
launch
wait (4 times, until buoy appears)
take buoy
go east
leave boat
take shovel
drop garlic
take shovel
drop buoy
take shovel
go northeast
```

```
dig
sand
dig sand
dig sand
dig sand
take scarab
drop shovel
go southwest
open buoy
take emerald
take garlic
go south
go south
cross rainbow
turn off lamp
go southwest
go up
go up
go northwest
go west
go west
go west
```

After returning to the house, switch the lamp on before entering any dark passage. Bank the trunk, trident, emerald, and scarab. Collect the jeweled egg from the tree, return to the Living Room, and prepare for the maze.

06. The egg, maze, and thief

Use the Cyclops name to open the shortcut, enter the treasure room from its stair, and give the egg to distract the thief. Bank the coins before returning with the knife. Combat varies; keep attacking until the thief dies and the egg reappears with its canary.

```
go east
go east
go north
go north
climb tree
take egg
climb down
go south
go east
enter house
go west
drop garlic
turn on lamp
go down
go north
go west
go south
go east
go up
take coins
take key
go southwest
go east
go south
go southeast
Ulysses
go up
give egg to thief
go down
go east
go east
put coins in case
take knife
go west
go west
go up
kill thief with knife (repeat until dead)
drop knife
take egg
take chalice
go down
go northwest
go south
go west
go up
go down
```

```
go northeast
unlock grate
open grate
go up
go south
climb tree
wind up canary
go down
take bauble
go south
go east
enter house
go west
put bauble in case
put chalice in case
take canary from egg
put canary in case
put egg in case
take garlic
```

07. Recover the bracelet, bar, map, and ending

From the Living Room, revisit the temple cave and rub the mirror to reach the mine branch. Pick up garlic at the Shaft Room before crossing the bat room again. Deposit the bracelet, then return to the quieted Loud Room for the final scored treasure.

```
go down
go north
go east
go east
go southeast
go east
go down
go south
go south
go down
go north
rub mirror
go north
go west
go north
go west
go north
go east
drop garlic
go north
go down
take bracelet
go up
go south
take garlic
go west
go south
```

```
go east
go south
go down
go up
put bracelet in case
go down
go north
go east
go east
go east
echo
take bar
go west
go west
go west
go south
go up
put bar in case
look
take map
read map
score
go east
go east
go south
go west
go southwest
enter barrow
```

Completion check: the trophy case should contain the full treasure set and the map. The final score should read 350/350. Entering the Stone Barrow displays the trilogy handoff and halts the story.

Executed opcode inventory

59 unique opcodes were observed across 417,780 instructions in the fresh-game route. Counts include all calls and story output. Opcode values are shown in hexadecimal; the form disambiguates opcode encodings. PCs are byte addresses.

Form	Opcode	Mnemonic	Count	First seq.	First PC
Variable	0x00	call	29,199	1	0x004F05
Long	0x14	add	14,575	3	0x005491
Long	0x01	je	42,724	5	0x005499
Long	0x15	sub	3,945	6	0x00549D
Short	0x00	jz	77,749	7	0x0054A1
Variable	0x01	storew	11,159	9	0x0054AC
Short	0x0B	ret	3,420	10	0x0054B1
Long	0x0F	loadw	24,531	19	0x0054B3
Short	0x0C	jump	24,787	22	0x0054C1
Variable	0x03	put_prop	38	113	0x004F3A
Long	0x0D	store	29,992	126	0x004F75
Long	0x0A	test_attr	28,420	131	0x004F7E
Short	0x02	print	1,010	133	0x006EE3
Short	0x0B	new_line	1,155	135	0x006F39
Long	0x10	loadb	8,827	136	0x006F3A
Long	0x09	and	1,900	137	0x006F3E
Variable	0x09	and	1	141	0x006F8C
Variable	0x06	print_num	11	142	0x006F92
Long	0x05	inc_chk	10,170	144	0x006FA4
Variable	0x05	print_char	34	146	0x006FAF
Short	0x00	rtrue	4,067	171	0x006FB6
Long	0x0E	insert_obj	674	176	0x004F91
Variable	0x08	push	3,950	180	0x008CA6
Variable	0x09	pull	471	182	0x008CAF
Long	0x0B	set_attr	749	185	0x008CE8
Long	0x06	jin	2,301	188	0x008CF5
Short	0x0A	print_obj	530	189	0x008CF9
Short	0x03	get_parent	4,439	190	0x008CFB
Long	0x11	get_prop	5,326	196	0x008D16
Short	0x02	get_child	9,371	207	0x008D54
Variable	0x01	je	11,871	221	0x008EDC
Short	0x01	get_sibling	10,392	226	0x008F3E
Short	0x05	inc	2,032	279	0x008F9B
Long	0x02	jl	4,254	281	0x008FA0
Short	0x01	rfalse	5,558	302	0x0050D5
Short	0x08	ret_popped	3,730	320	0x008D70
Variable	0x04	sread	405	387	0x00590C
Long	0x04	dec_chk	2,183	394	0x005932
Long	0x07	test	5,586	403	0x005BB9
Long	0x03	jg	6,605	408	0x005BBD
Long	0x16	mul	1,056	419	0x005A38

Form	Opcode	Mnemonic	Count	First seq.	First PC
Variable	0x02	storeb	807	422	0x005A44
Variable	0x0D	store	637	624	0x00672B
Long	0x0C	clear_attr	2,960	628	0x006742
Long	0x12	get_prop_addr	8,675	635	0x006A65
Short	0x04	get_prop_len	2,317	650	0x006D30
Long	0x17	div	2,074	651	0x006D33
Short	0x03	print_ret	166	866	0x0080F4
Variable	0x05	inc_chk	400	1,037	0x0054EC
Short	0x0D	print_paddr	333	1,665	0x008417
Long	0x13	get_next_prop	39	22,705	0x010127
Variable	0x07	random	122	25,500	0x00A960
Short	0x09	remove_obj	8	28,299	0x0078CC
Short	0x0E	load	24	35,427	0x008B09
Short	0x06	dec	2	98,965	0x00DD55
Short	0x07	print_addr	7	242,576	0x00639B
Variable	0x03	jg	5	287,809	0x0105E6
Variable	0x02	jl	6	312,553	0x00FBB0
Short	0x0A	quit	1	417,780	0x006E07

Replay completion: Halted; route hash 3F81D0A8B88AB1F9B3D24D9C4B3575D54663547EA47BA310DD0A7168D10E9446. The full deterministic route and compressed JSONL trace are stored locally under .agent/regression-routes/ and are intentionally ignored by Git.